A DERIVED COMPLETE MODULE WHOSE REDUCTION MODULO THE IDEAL IS FINITE IS THE COMPLETION OF A FINITE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span names A^wedge without defining it; the definition is supplied inside the counted proof span, which is delivered verbatim. The four references inside the counted proof span (including the two Algebra chapter references) are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The naive completion of a Noetherian ring agrees with its derived completion. invoked at proof lines 27302-27305. Identifies the derived completion of A with the usual completion A^wedge.
- A derived complete module annihilated by I is zero. invoked at proof lines 27307-27308. Shows the map q is surjective, giving finite generation over A^wedge.
- Completeness of finite modules over a complete Noetherian ring, and Artin-Rees. invoked at proof lines 27310-27314. Deduces that M is I-adically complete.

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05GH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7061–7061. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 031B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3978–3978. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 26100: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26100–26100. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 27218: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27218–27218. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be an ideal of a Noetherian ring A. Let M be a derived complete A-module. If M/IM is a finite A/I-module, then $M = \lim M/I^nM$ and M is a finite $A^\wedge$-module.*

Proof. Assume M/IM is finite. Pick $x_1, \ldots, x_t \in M$ which map to generators of M/IM. We obtain a map $A^{\oplus t} \to M$ mapping the ith basis vector to x_i. By Proposition 27218 the derived completion of A is $A^\wedge = \lim A/I^n$. As M is derived complete, we see that our map factors through a map $q : (A^\wedge)^{\oplus t} \to M$. The module $\mathrm{Coker}(q)$ is zero by Lemma 26100. Thus M is a finite $A^\wedge$-module. Since $A^\wedge$ is Noetherian and complete with respect to $IA^\wedge$, it follows that M is I-adically complete (use Algebra, Lemmas 05GH, 031B, and 00IN). □